

Giorgio Caizzi

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 [giorgio-caizzi](#)  [giocaizzi](#)

Engineer who owns systems end to end — event-driven backends, observability and IaC, with AI on top. Self-taught into code at AECOM, now a hands-on tech lead shipping production systems for banks, media and retail.

EXPERIENCE

Associate Manager, Products (prev. Senior Consultant)

May 2024 – Present

Jakala S.p.A.

Milan, Italy

- Built a production RAG over 600,000 pages of internal banking regulation (PDF, Word, Excel) as lead engineer on a year-long embedded Banca Generali engagement — structure-aware chunking, hierarchical retrieval, version-aware metadata and an event-driven sync pipeline into the bank's platforms, quality-gated by a golden evaluation set. In production, rolling out across 1,600 employees.
- Built a governed tool layer that lets AI agents run SQL and geospatial operations — a GraphQL wrapper with RBAC constraining each agent call to the user's role. Non-technical users chat with their data beside a live map; productized for clients including Regione Lombardia and NielsenIQ.
- Led the engineering team that built an AI pipeline generating on-model product photography for Benetton (ComfyUI, Flux, LoRAs on Azure GPUs) at up to 90 SKUs/day with hybrid human-and-AI QA — replacing studio photoshoots for most of the brand's ecommerce imagery by autumn 2025.
- Developed the vision-language pipeline that estimates a meal's carbohydrates from a single photo for T.Conto (a consumer nutrition app live on the App Store and Google Play), using agents to retrieve ingredient reference values from a vector store of the official food-composition tables.

Environmental Engineer

Feb. 2021 – May 2024

AECOM URS Italia S.p.A.

Milan, Italy

Remediation of contaminated industrial sites for ENI, Q8 and the US DoD.

- Built Python pipelines (geopandas, GDAL, rasterio, PostGIS) that turned days of manual GIS work into hours — open-sourced on GitHub as [py3dinterpolations](#).

Brand Activation & Live Events

Nov. 2013 – Feb. 2021

Red Bull, GoPro EMEA

Italy & Europe

Seven years of live brand-activation at major sporting events across Europe, alongside my studies — public speaking and running games/giveaways for crowds, trilingually.

EDUCATION

MSc (102/110) and BSc (97/110) in Environmental & Land Planning Engineering

Milan, Italy

Politecnico di Milano

Sep. 2014 – Dec. 2020

Relevant coursework: computer science (30/30), GIS & spatial data, statistical models & stochastic processes, remote sensing

Theses: PM2.5 spatial-interpolation algorithm (Wiseair S.r.l.); facility-location optimization for Milan's public water network (MM S.p.A.)

The British School of Brussels

Brussels, Belgium

International Baccalaureate Diploma

Sep. 2012 – May 2014

TECHNICAL SKILLS

GenAI & LLMops: RAG & retrieval engineering, LLMs & multimodal, agentic orchestration & MCP, evaluation & guardrailing, AI observability, OpenAI, Anthropic & Azure OpenAI

Backend & Systems: Python & TypeScript, FastAPI, REST & GraphQL, event-driven pipelines & queues, OpenTelemetry distributed tracing, Docker, CI/CD, git

Data: SQL & NoSQL, Redis, vector & graph stores

Cloud & Infrastructure: Azure, GCP, Terraform (IaC), networking, serverless

Geospatial: PostGIS, GDAL, geopandas, shapely, rasterio, ArcGIS & QGIS

PROJECTS

-  **ralph-copilot** autonomous agent loop for GitHub Copilot ★138
-  **greenhouse** smart-irrigation app exposing an MCP server to LLM agents
-  **btc-cycles** bitcoin-cycle analytics with a live daily dashboard

Copilot custom agents, git-as-memory
Python, FastAPI, MCP, Docker
Python, PyPI, CI/CD, plotly

PERSONAL SKILLS

Languages: Italian (native), English (bilingual), French (professional working proficiency).

Interests: Surfing, skiing and road-cycling, chased worldwide: Indonesia, Angola, South Africa, across Europe.